THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS TWO-CHAPTER TEST SET D

Chapters 11 & 12: **Areas Related to Circles, Surface Areas and Volumes** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Use $\pi = \frac{22}{7}$ unless stated otherwise.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The area of a quadrant of a circle of radius 21 cm is: 1

(a) 693 cm² **(b)** 173.25 cm² **(c)** 346.5 cm² **(d)** 1386 cm²

2. The arc length of a quadrant of radius 28 cm is: 1

(a) 176 cm **(b)** 44 cm **(c)** 22 cm **(d)** 88 cm

3. The area swept by a 14 cm minute hand in 30 minutes is: 1

(a) 616 cm² **(b)** 44 cm² **(c)** 308 cm² **(d)** 154 cm²

4. A student states: “A sector of angle 180° is a semicircle.” The correct response is: 1

(a) Correct **(b)** It is a quadrant

(c) Incorrect **(d)** Cannot be determined

5. For the perimeter of a quadrant of radius 14 cm, Student A says 50 cm; Student B says 22 cm. Then: 1

(a) Student B is correct **(b)** Student A is correct

(c) Both are correct **(d)** Both are wrong

6. The volume of a cylinder of radius 2 cm and height 7 cm is: 1

(a) 28 cm³ **(b)** 176 cm³ **(c)** 44 cm³ **(d)** 88 cm³

7. The total surface area of a cone of radius 3 cm and slant height 5 cm is: 1

(a) 30π cm² **(b)** 24π cm² **(c)** 15π cm² **(d)** 9π cm²

8. The volume of a hemisphere of radius 7 cm is: 1

(a) 359.33 cm³ **(b)** 308 cm³ **(c)** 1437.33 cm³ **(d)** 718.67 cm³

9. A student states: “The volume of a cone is half that of a cylinder with the same base and height.” The correct response is: 1

(a) Incorrect — it is one-third **(b)** It is two-thirds

(c) Correct **(d)** Cannot be determined

10. Three unit cubes are placed in a row. Student A says the volume of the cuboid is 3 cubic units; Student B says 1. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the area of a sector of radius 3.5 cm with angle 90°. 2

12. The area of a sector of angle 90° is 38.5 cm². Find the radius. 2

13. A hemisphere of radius 3 cm is placed on one face of a cube of edge 6 cm. Find the total surface area. 2

14. A cylinder of radius 7 cm and height 4 cm has a cone of height 3 cm on top (same radius). Find the volume. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the area of the minor segment of a circle of radius 6 cm when the sector angle is 60°. (Use $\pi = 3.14$, $\sqrt{3} = 1.73$) 3

16. Two identical cones, each of radius 3 cm and height 4 cm, are joined base to base. Find the volume and surface area of the solid. 3

17. A minute hand 21 cm long moves from 6:05 to 6:40. Find the area swept. 3

OR

Find the perimeter and area of a major sector of radius 14 cm and angle 270°. 3

18. A tent is a cylinder of height 2 m and radius 3.5 m surmounted by a cone of slant height 5 m. Find the canvas needed and its cost at ₹100 per m². 3

OR

A pen holder is a cuboid 10 cm × 5 cm × 4 cm with three cylindrical holes, each of radius 0.7 cm and depth 3 cm. Find the volume of wood. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** A security camera covers a sector of radius 10 m and angle 72°. (Use $\pi = 3.14$) 5

(i) Find the area covered.
(ii) Find the arc length.
(iii) How many such cameras are needed to cover a full circle?
(iv) Find the area covered if the range is 20 m.

OR

A chord AB of a circle of radius 7 cm subtends 120° at the centre. Find 5

(i) the sector area
(ii) the triangle area
(iii) the minor segment
(iv) the major segment. (Use $\sqrt{3} = 1.73$)

20. **Case study:** A trophy has a cylindrical base of radius 5 cm and height 4 cm with a hemisphere of radius 5 cm on top. (Use $\pi = 3.14$) 5

(i) Find its volume.
(ii) Find the curved area to be gold-plated.
(iii) Find the cost of plating at ₹2 per cm².

OR

A solid is a cylinder of radius 7 cm and height 10 cm, with a cone of height 24 cm on top and a hemisphere at the bottom (same radius). Find 5

(i) its volume
(ii) its total surface area.

END OF QUESTION PAPER — **Total: 40 Marks**